



Indian Institute of Technology Kharagpur

Class Test I 2024-25

Date of Examination: 21 Jan, 2024

Duration: 30 Minutes

Subject No.: CS60050

Subject: Machine Learning

Department/Center/School: Computer Science

Credits: 3

Full marks: 30

Name: _____

Roll Number: _____

Question:	1	2	3	Total
Points:	10	10	10	30
Score:				

Write the answers in the space provided after the questions only.

- (a) (5 points) What primitive functions/operations are needed to implement the candidate elimination algorithm on an arbitrary hypothesis space?

Solution:

Finding all immediate generalizations / specializations of a given hypothesis.

Checking whether a given hypothesis h_1 is a generalization / specialization of another hypothesis h_2 .

Check whether a given hypothesis is consistent with a given example.

- (b) (5 points) Consider a decision stump as the hypothesis with input features x_1 and x_2 (both real numbers). A hypothesis is of the form:

for some feature x_i : if $(x_i \geq \tau_i)$ then predict +1 else predict -1

where τ_i is a real threshold. What is the size of the hypothesis space?

Consider a hypothesis in this space being learned using 10 unique training datapoints. What is the maximum size of the version space, considering that τ_i can only be chosen from one of the x_i values in the training dataset or $\pm\infty$?

Solution: Ans: Hypothesis space: infinite since τ_i can be anything.

Maximum size of version space, both the below answers are correct:

Considering the maximum size over all fixed labelings, it is: $4(2 + 2)$

Considering the maximum number of hypothesis which can be consistent with some labeling: $22(12 + 12)$

2. (10 points) Consider the weighted version of a simple least squares linear regression problem: where α_i denotes the weightage of the i^{th} training datapoint: (x_i, t_i) , for $i = 1, \dots, N$. The predictor is $y(x) = wx$ and the loss function is $l(w) = \sum_{i=1}^N \alpha_i (t_i - y(x_i))^2$. Write the formula for $w^* = \arg \min_w l(w)$.

Solution:

$$l(w) = \sum_{i=1}^N \alpha_i (t_i - wx_i)^2$$

$$\text{Expanded loss function: } l(w) = \sum_{i=1}^N \alpha_i t_i^2 - 2w \sum_{i=1}^N \alpha_i t_i x_i + w^2 \sum_{i=1}^N \alpha_i x_i^2$$

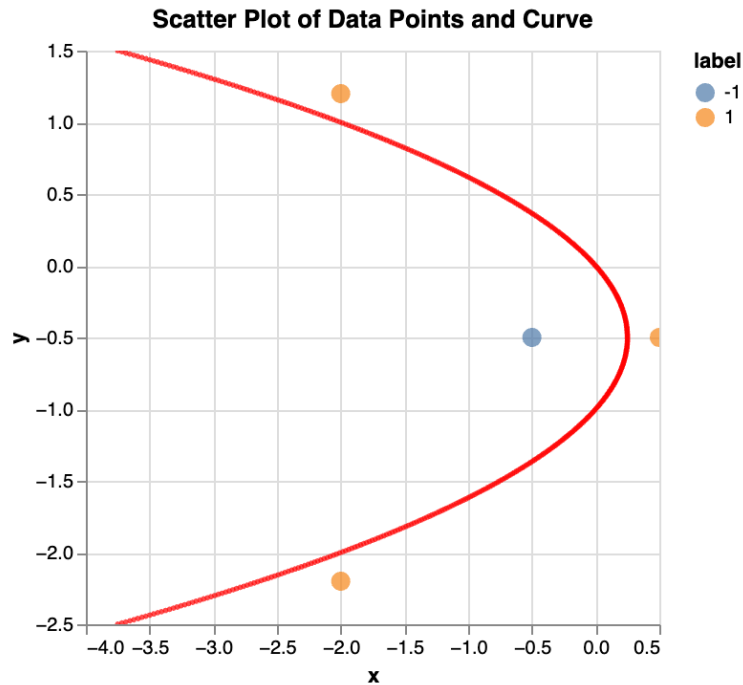
$$\text{Derivative of the loss function: } \frac{\partial l(w)}{\partial w} = -2 \sum_{i=1}^N \alpha_i t_i x_i + 2w \sum_{i=1}^N \alpha_i x_i^2$$

$$\text{Setting the derivative to zero: } w \sum_{i=1}^N \alpha_i x_i^2 = \sum_{i=1}^N \alpha_i t_i x_i$$

$$\text{Optimal } w^*: w^* = \frac{\sum_{i=1}^N \alpha_i t_i x_i}{\sum_{i=1}^N \alpha_i x_i^2}$$

3. (10 points) Consider the following binary classification dataset of the format $((x_1, x_2), y)$: $\{((-0.5, -0.5), -1), ((0.5, -0.5), +1), ((-2, -2.2), +1), ((-2, 1.2), +1)\}$ Is this linearly separable? Write the lowest order polynomial feature map which makes this dataset linearly separable. Write the coefficients of each feature such that the dataset is separated.
Hint: plot the datapoints and come up with a curve equation separating the classes.

Solution:



Equation of the separator: $x_1 + x_2 + x_2^2 = 0$

Rough space

